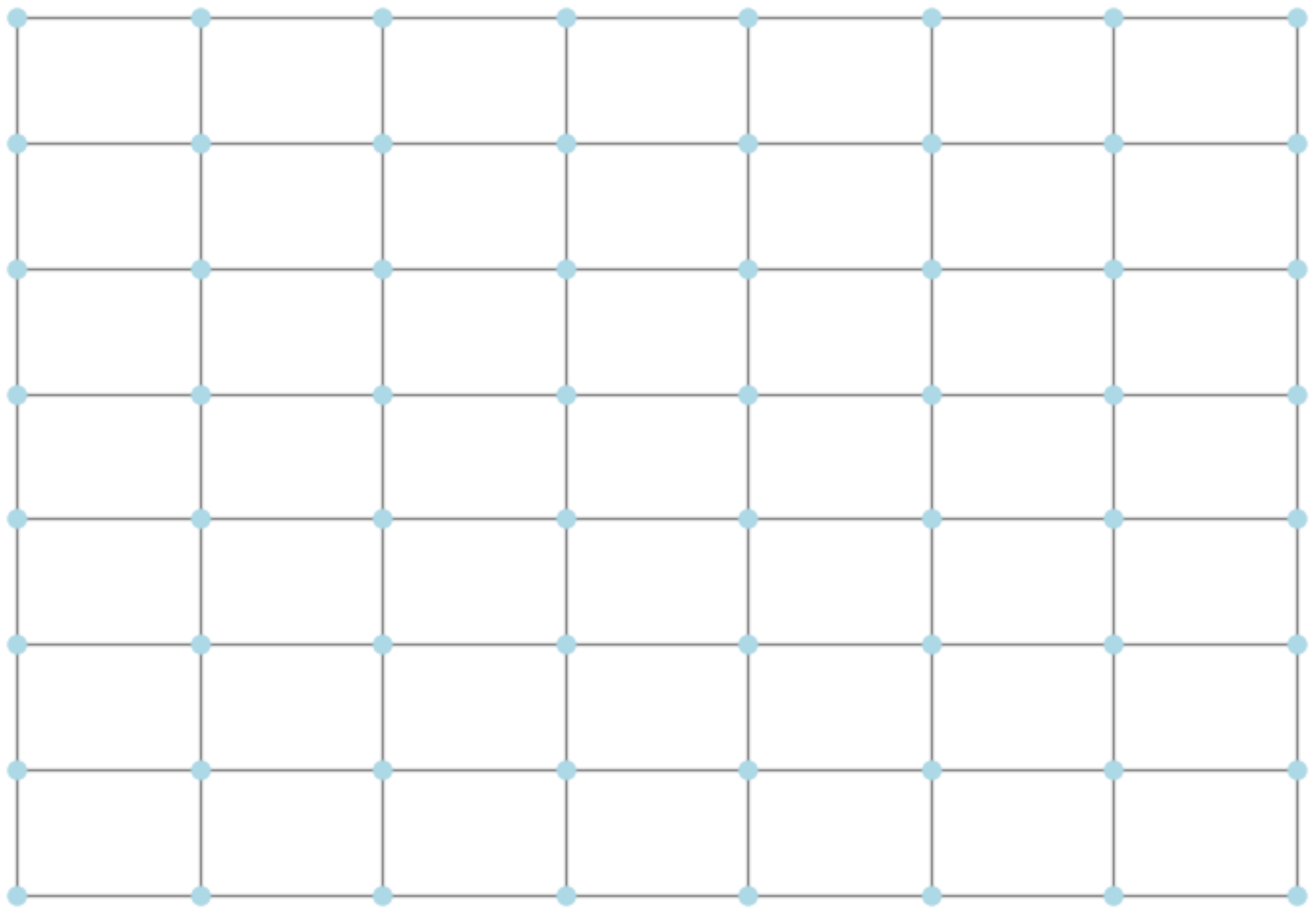
Complex Network Basics with Python NetworkX

Reference:
Chapter 15, Introduction to the modelling and analysis of complex systems, by Hiroki Sayama
Chapter 6, Networks, by Mark Newman

What is a COMPLEX network?

Simple/regular



Complex

https://en.wikipedia.org/wiki/Network_theory

Definitions
- Graph theory

A **network** (or **graph**) consists of a set of **nodes** (or **vertices**) and a set of **edges** (or **links**) that connect those nodes.

Node j is called a **neighbor** of node i if (and only if) node i is **connected** to node j.

Adjacency matrix A matrix with rows and columns labeled by nodes, whose i-th row, j-th column component a_{ij} is 1 if node i is a neighbor of node j, or 0 otherwise.
Adjacency list A list of lists that lists out all the neighbours of each node

Degree The number of edges connected to a node. Node i's degree is often written as deg(i) or k_i.

Path A sequence of edges that form a continuous route between two nodes.

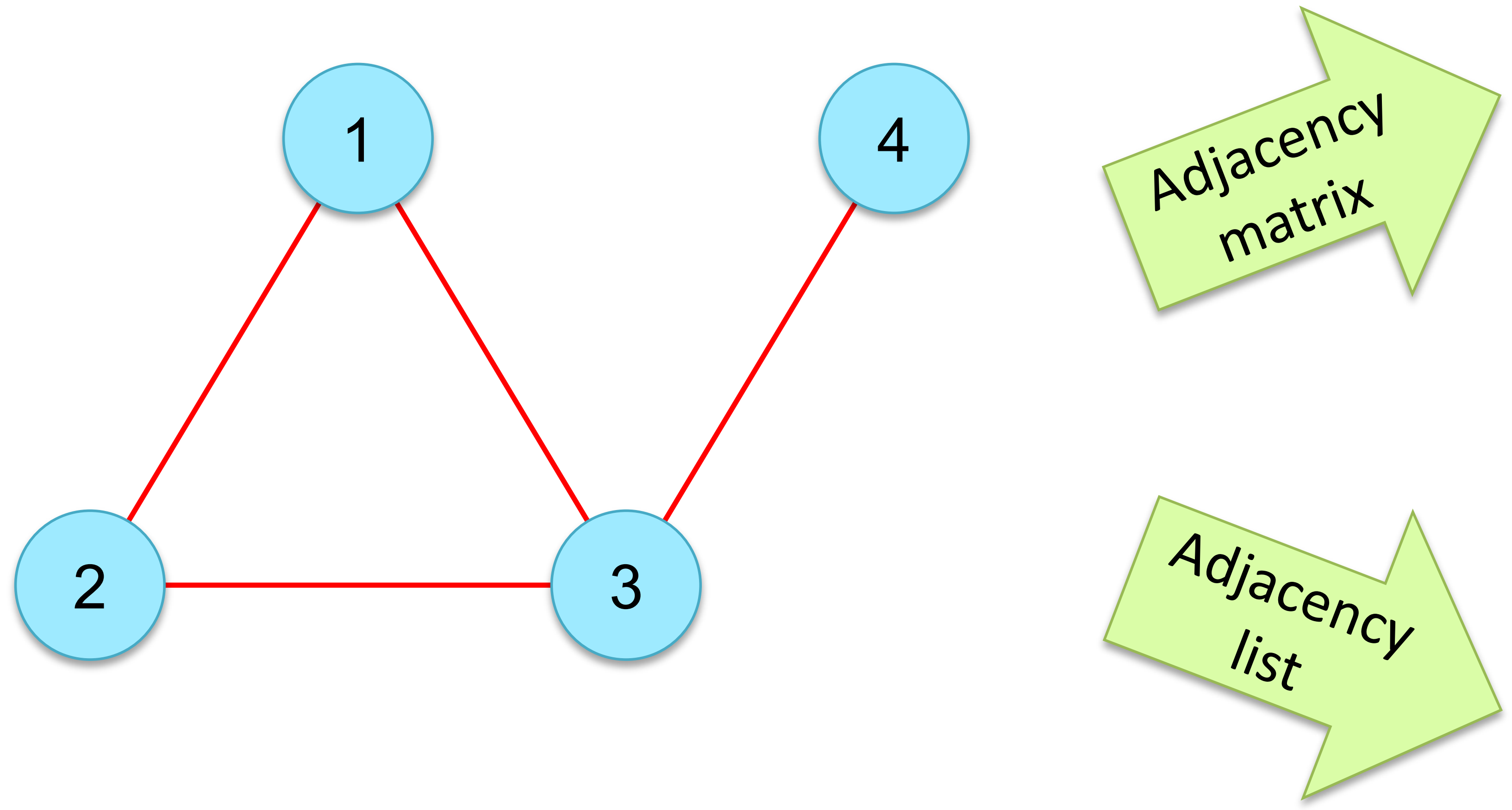
Cycle A sequence of edges that connects a node back to itself.

Subgraph: Part of a graph

Connected graph: A graph in which a path exists between any pair of nodes

Connected component: A subgraph that is connected within itself but not to the rest of the graph

Network (graph)



$$\begin{array}{c} i: \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} j:\ 1 \quad 2 \quad 3 \quad 4 \\ \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \end{array}$$

i :

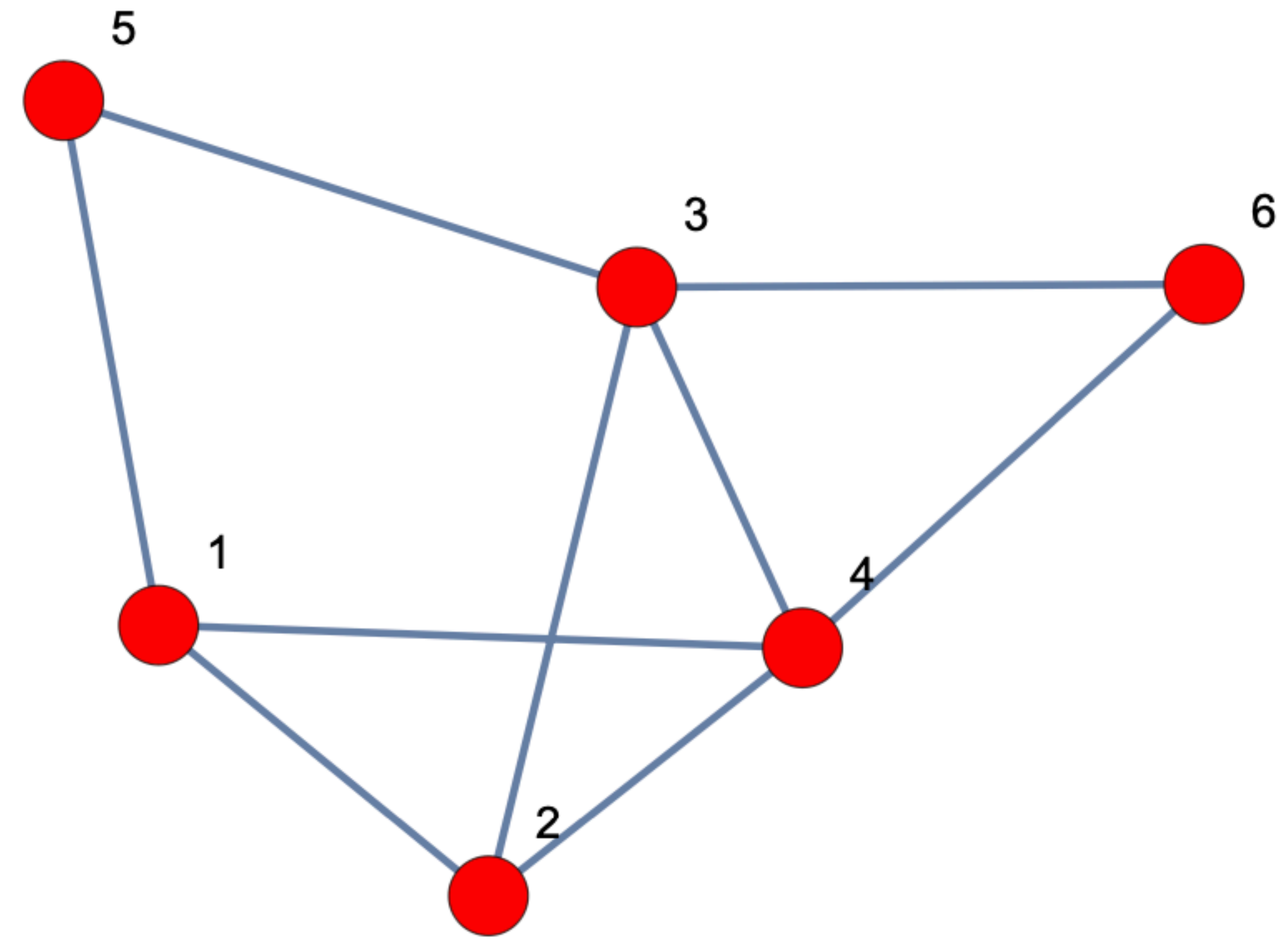
$1 \rightarrow \{2, 3\}$

$2 \rightarrow \{1, 3\}$

$3 \rightarrow \{1, 2, 4\}$

$4 \rightarrow \{3\}$

Short exercise



1. Represent the network as
   (a) an adjacency matrix.
   (b) an adjacency list
2. Determine the degree of each node
3. Classify the following as path, cycle or neither
   - 6-3-2-4-5
   - 1-4-6-3-2
   - 5-1-2-3-5
4. Identify all connected 3-node subgraphs
5. Remove node 3 and 4. Then identify the connected components in the graph.

Definitions
- Small note on graph with multiple components

"The adjacency matrix of a network with more than one component can be written in block diagonal form, meaning that the non-zero elements of the matrix are confined to square blocks along the diagonal of the matrix, with all other elements being zero:"

$$\mathbf{A} = \begin{pmatrix} \Box & 0 & \cdots \\ 0 & \Box & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}$$

Definitions
- graph theory

Complete graph: A graph in which any pair of nodes are connected

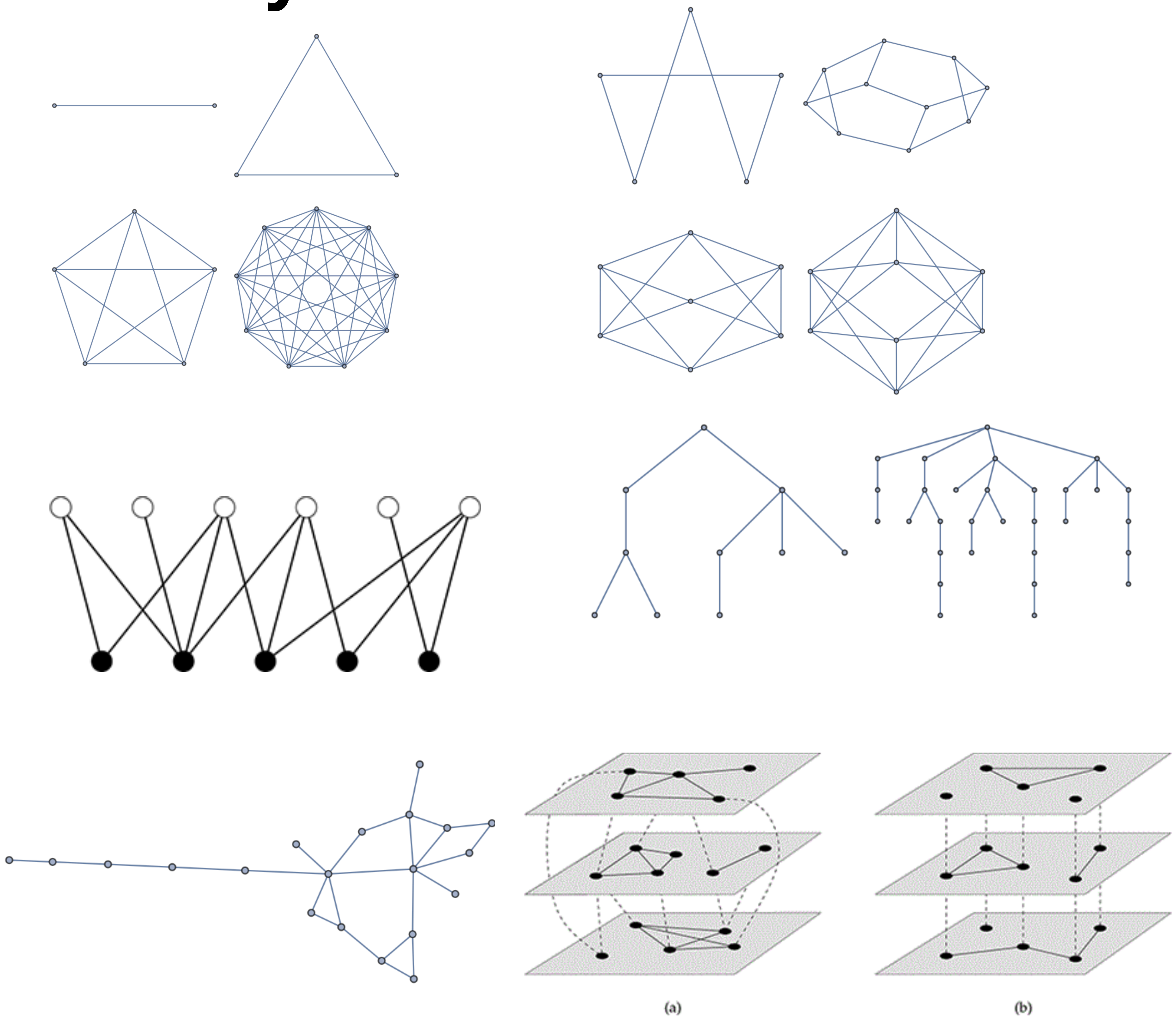
Regular graph: A graph in which all nodes have the same degree

Bipartite graph: A graph whose nodes can be divided into two groups so that no edges connects nodes within each group.

Tree graph: A graph that has no cycle

Planar graph: A graph that **can** be drawn on a plane without having any edges cross.

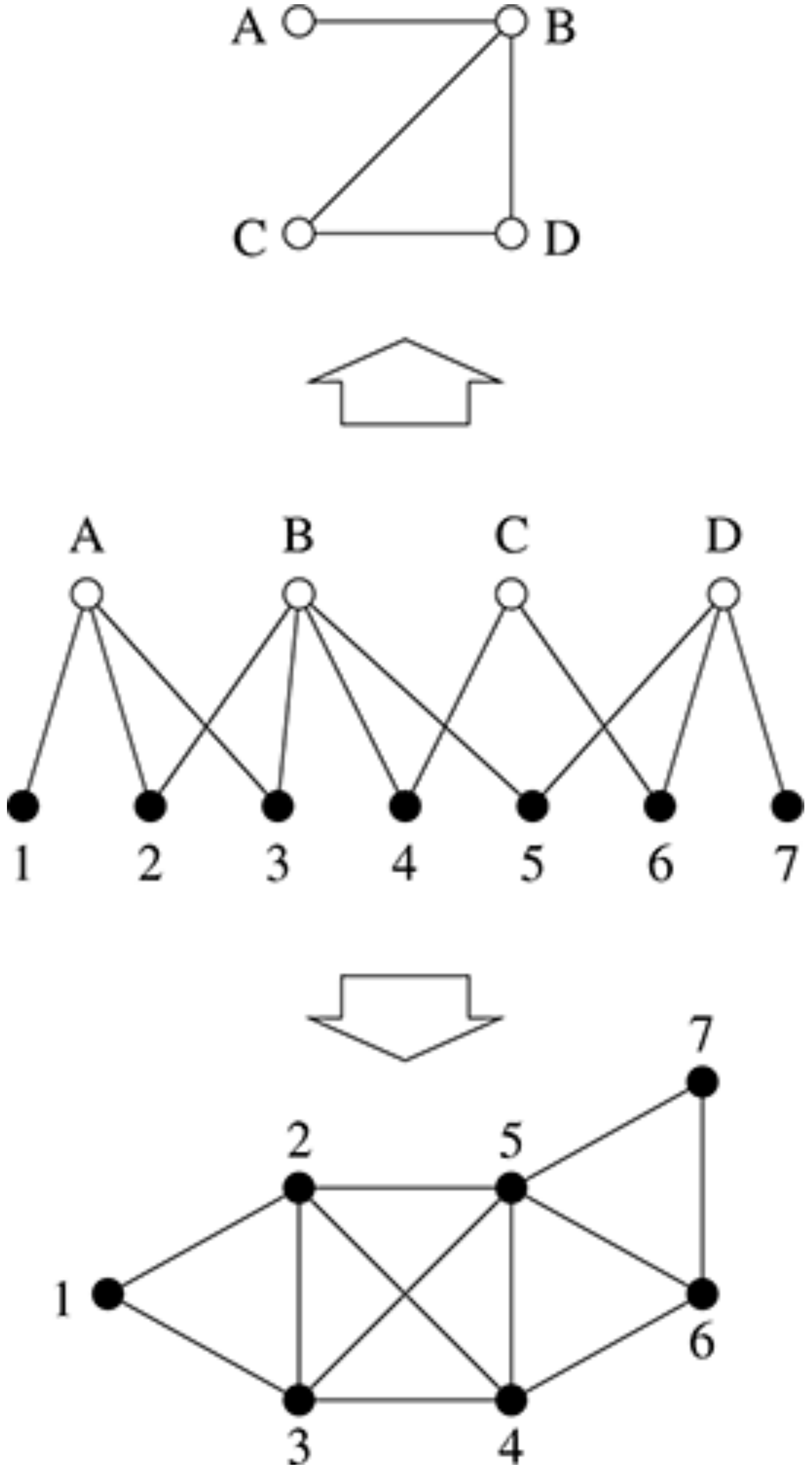
Multi-layer graph: A set of individual graphs, each representing nodes and edges of a particular type, plus inter-layer edges between networks
Multiplex graph: A special case of multi-layer network in which the nodes represent the same set of objects or people in each layer.



Short exercise

1. What is the number of edges in a regular graph with n nodes each with degree k?
2. What is the number of edges in a tree graph with n nodes?
3. Give real life examples for each of the graph types:
   1. Complete graph
   2. Regular graph
   3. Bipartite graph
   4. Tree graph
   5. Planar graph
   6. Multi-layer graph
   7. Multiplex graph

More on bipartite network



Bipartite networks (graphs) can be also represented as other network structures, by focusing on only one side of the network

Question: what is the adjacency matrix for each form of the network?

Practical machine learning problem: Netflix movie recommender system.

https://medium.com/@s-kp/machine-learning-case-study-516ac43accda

4				5	
4	5		5		
	5		4		3

Historical data

Definitions - More

Undirected edge: A symmetric connection between nodes. (friendship)

Undirected graph: A graph made with undirected edges. (Facebook)

Directed edge: An asymmetric connection between nodes. (Follower)

Directed graph: A graph made with directed edges. (Twitter/X)

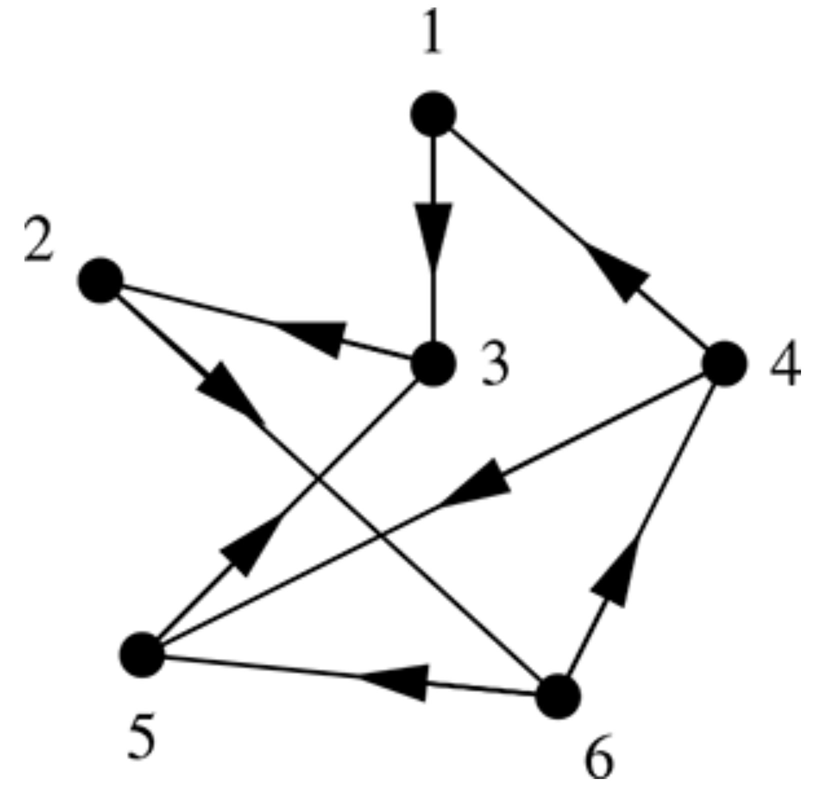
Unweighted edge: An edge without and weight values. Two nodes are either connected or not.

Weighted edge: An edge that has a weight value(s). Can represent strength, distance, etc.

Question: How to write down the adjacency matrix for:

1. Directed graph
2. Weighted graph

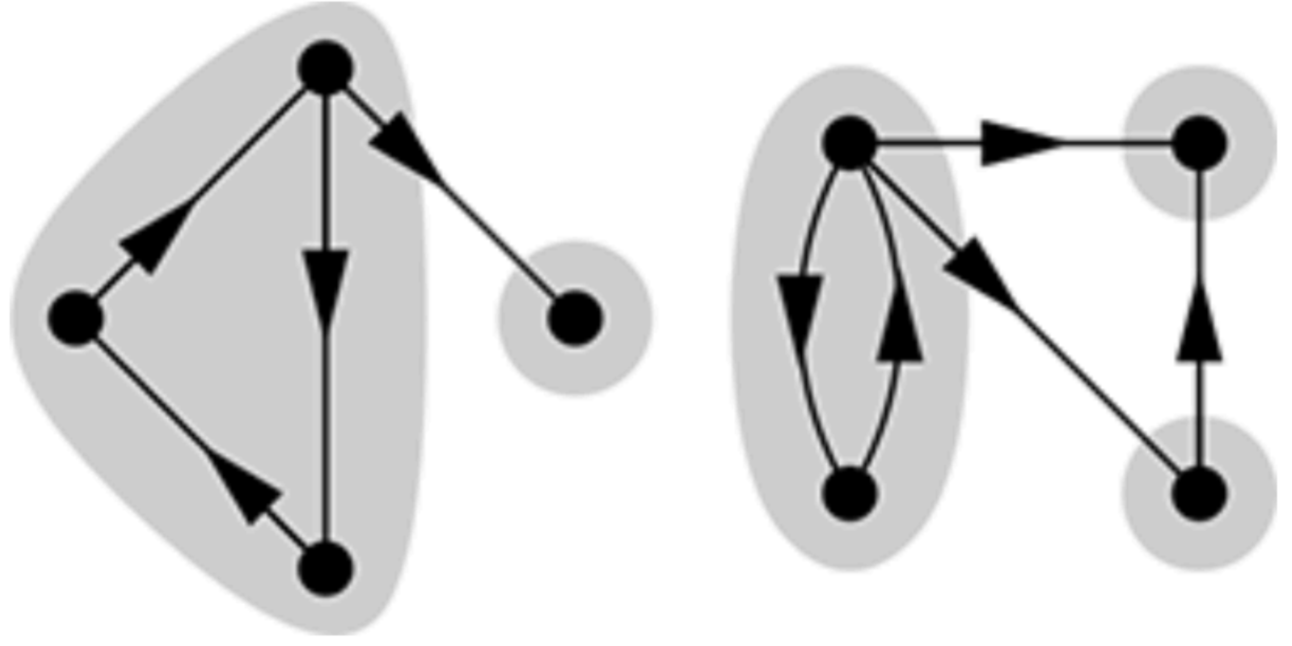
Question: What is the difference between the adjacency matrix of directed and undirected graph?



Definitions
- Directed components

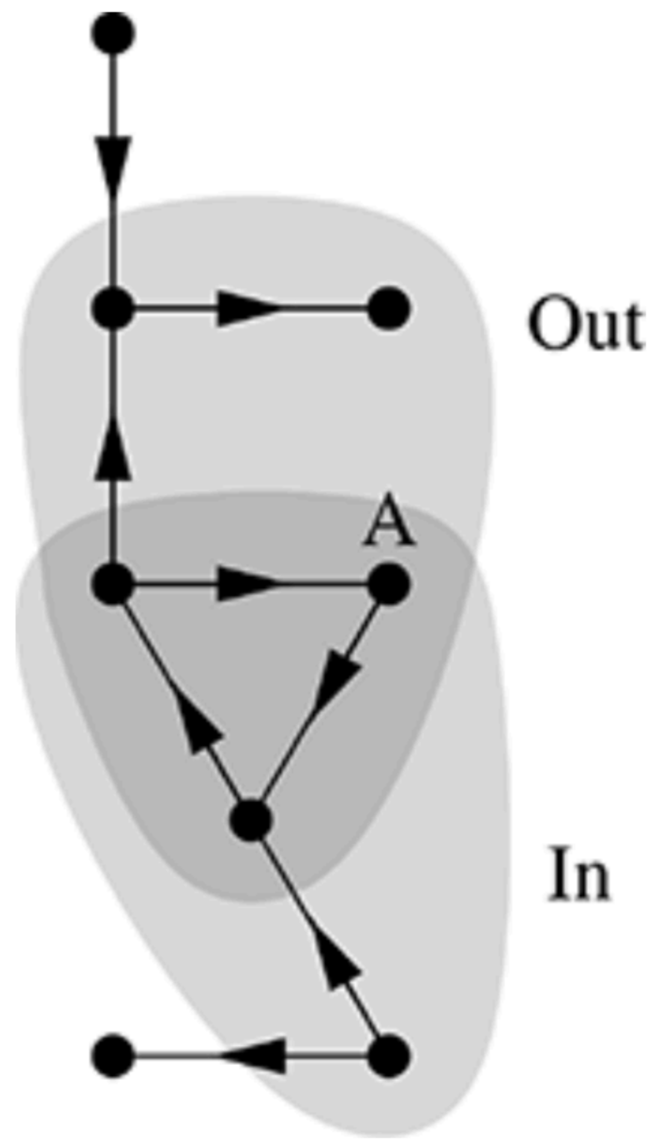
Strongly connected component: A subgraph such that there is a path between every pair of nodes in both directions.

Weakly connected component: A subgraph such that there is a path between every pair of nodes if links are undirected.



Out component: the set of nodes that are reachable via directed paths starting from a specified node A, including A.

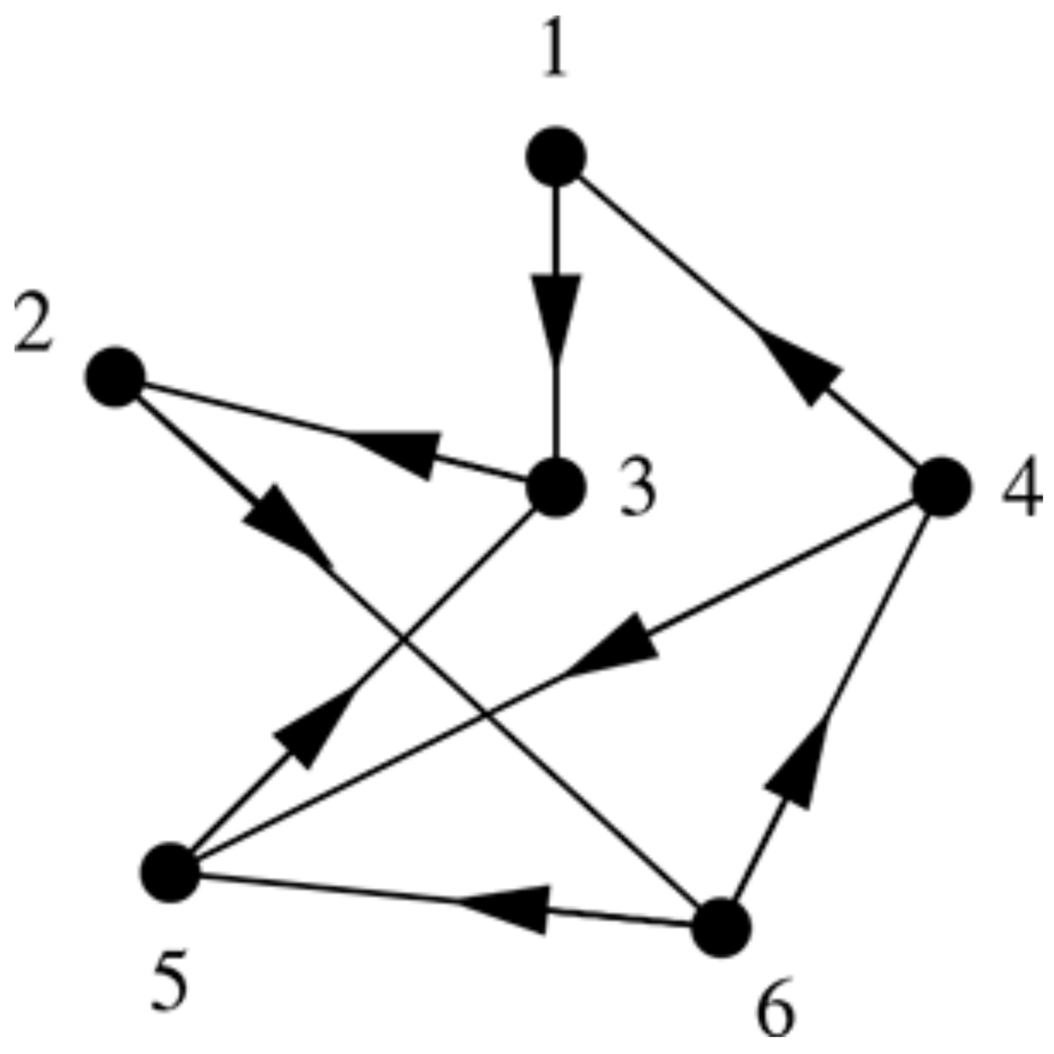
In component: the set of nodes that via directed paths can reach a specified node A, including A.



Definitions
- Adjacency matrix and degree

Adjacency matrix of the undirected network

$$\begin{pmatrix} 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \end{pmatrix}$$



Adjacency matrix of the directed network

$$\begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{pmatrix}$$

Question: How to calculate the degree of each node from the adjacency matrix?

Answer: Undirected network: Sum up each row or column
directed network: out degree - sum up each row; in degree - sum up each column

Question: How to find the 2nd nearest neighbor?

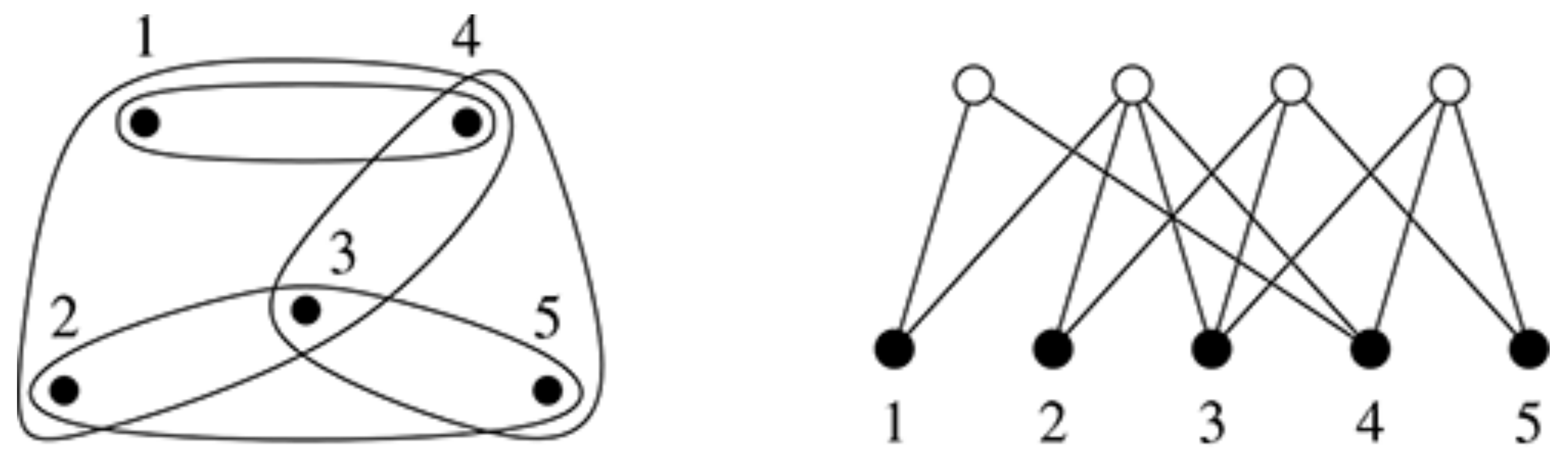
Answer: A^2

Definitions
- Hypergraph

Hyperedge: An edge that can connect more than 2 nodes at a time.

Hypergraph: An graph that is made of hyper edges.

A hyper graph can be represented a bipartite network.



(a) (b)

Network	Node	Group
Film actors	Actor	Cast of a film
Coauthorship	Author	Authors of an article
Boards of directors	Director	Board of a company
Social events	People	Participants at social event
Recommender system	People	Those who like a book, film, etc.
Keyword index	Keywords	Pages where words appear
Rail connections	Stations	Train routes
Metabolic reactions	Metabolites	Participants in a reaction

Definitions
- Sparsity and density

Density: the fraction of edges that are present in the network over all possible edges

$$\boxed{\rho = \frac{m}{\binom{n}{2}} = \frac{2m}{n\,(n-1)} = \frac{c}{n-1} \approx \frac{c}{n}}$$

n is large

m is the total number of links

$c = 2m/n$ is the average degree

In many if not most big real world networks, the density is very low (very sparse).
Hence, adjacency matrix is very inefficient computationally - requiring $\sim n^2$elements;
adjacency list is much more efficient — requiring the order of n elements only.

Adjacency matrix -> adjacency list:
huge reduction in computational complexity for large n

Practice

Coding practice with NetworkX